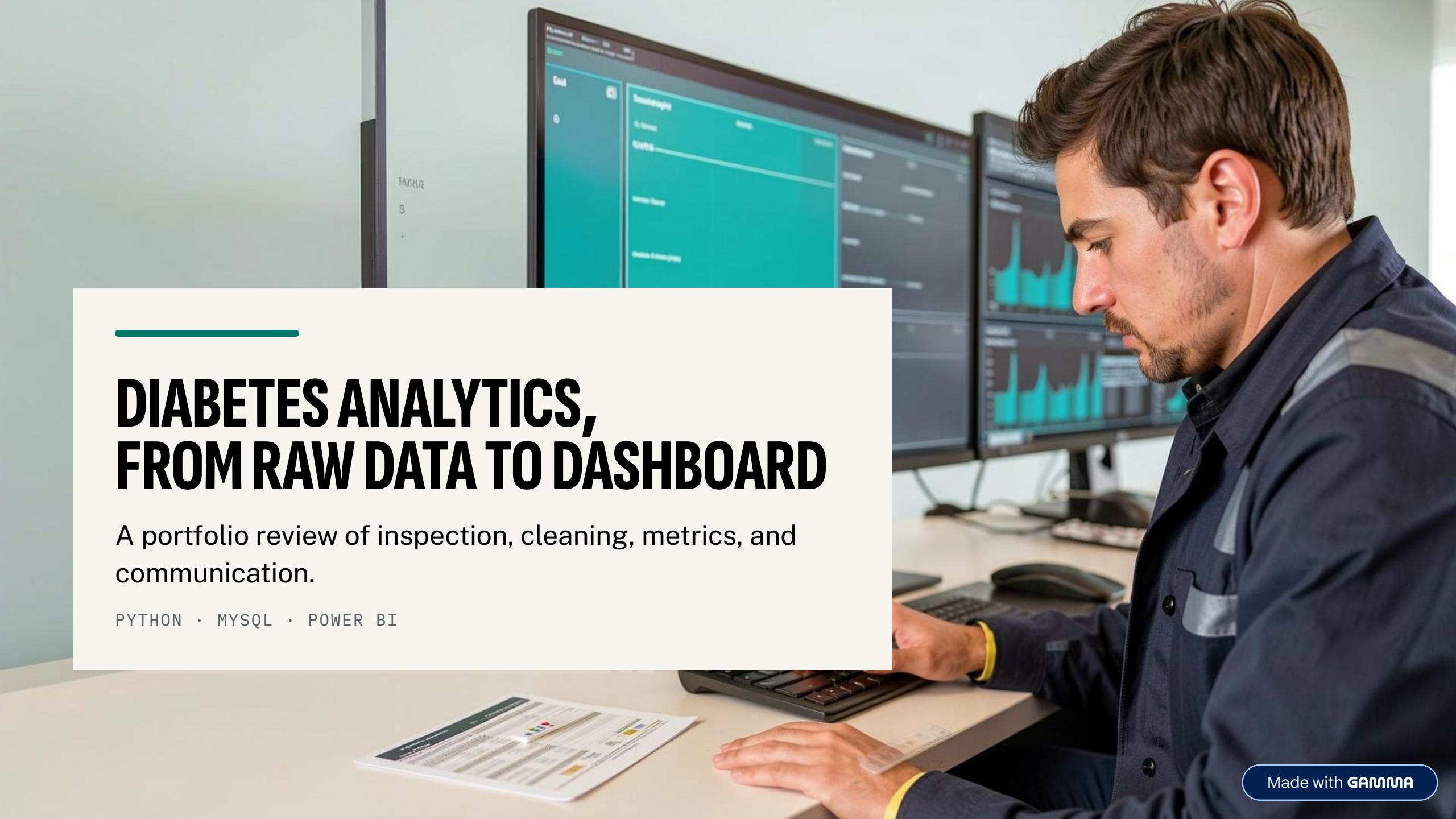


DIABETES ANALYTICS, FROM RAW DATA TO DASHBOARD

A portfolio review of inspection, cleaning, metrics, and communication.

PYTHON · MYSQL · POWER BI

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THREE TOOLS. ONE EVIDENCE CHAIN.

END-TO-END WORKFLOW

AN END-TO-END WORKFLOW ACROSS THREE TOOLS



Python

Inspection, cleaning, and exploratory analysis



MySQL

Grouped metrics and validation queries



Power BI

Interactive reporting and dashboard communication

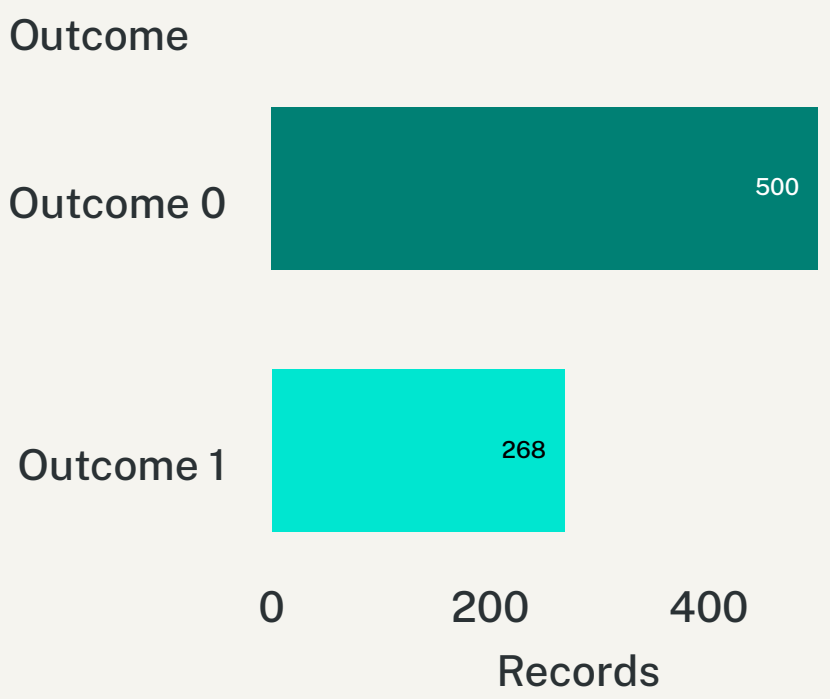
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DATASET PROFILE

THE DATASET IS COMPACT, STRUCTURED, AND OUTCOME-LABELED

RECORDS	OUTCOME 0	OUTCOME 1
768	500	268
Across nine columns	65.1%	34.9%

INPUTS: PREGNANCIES, GLUCOSE, BLOOD PRESSURE, SKIN THICKNESS, INSULIN, BMI, PEDIGREE SCORE, AND AGE.



Outcome 0 is the larger class; Outcome 1 is about one-third of records.

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DATA QUALITY

ZERO VALUES WERE TREATED AS LIKELY MISSING MEASUREMENTS

The check focused on Glucose, BloodPressure, SkinThickness, Insulin, and BMI.

GLUCOSE

5

Original zeros

BLOODPRESSURE

35

Original zeros

SKINTHICKNESS

227

Original zeros

INSULIN

374

Original zeros

BMI

11

Original zeros

MEANINGFUL ZEROES

PREGNANCIES AND OUTCOME ZEROS STAYED MEANINGFUL.

Not every zero is missing data. The field's meaning determines the treatment.

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COMPLETE THE WORKING TABLE.

CLEANING LOGIC

MEDIAN IMPUTATION MADE THE WORKING TABLE COMPLETE

01

Replace selected zero measurements

Convert likely missing measurements to missing values.

02

Fill with each column’s median

Use the median to complete the working table.

03

Confirm zero remaining missing values

Validate the cleaned result before analysis.

Insulin and SkinThickness still need explicit limitation language because their missing-value burden is high.

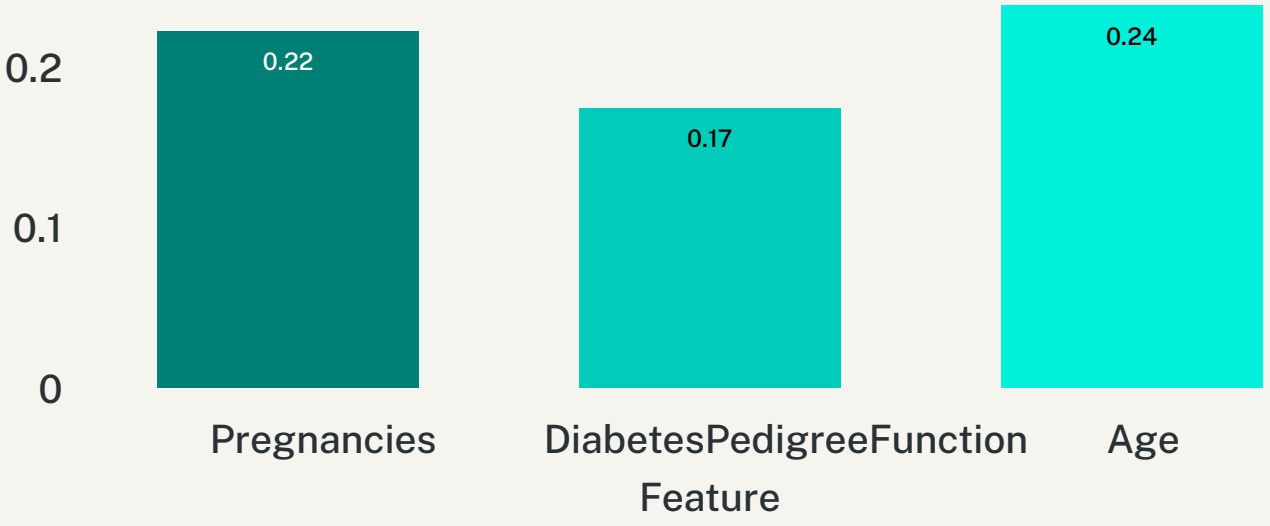
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EXPLORATORY ANALYSIS

GLUCOSE IS THE CLEAREST DESCRIPTIVE SEPARATION

The notebook’s box plot shows generally higher glucose measurements among Outcome 1 records.

Recorded correlation with Outco...



RECORDED CORRELATIONS

ASSOCIATION IS DESCRIPTIVE, NOT CAUSAL.

Pregnancies 0.222, DiabetesPedigreeFunction 0.174, and Age 0.238.

These are associations in this dataset, not causation or clinical risk assessment.

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SQL TRANSLATES THE EXPLORATION INTO REPEATABLE METRICS

The handoff from notebook insight to reusable query is where analysis becomes operational.

The next queries should test both the cleaned values and the logic behind each metric.

SQL OUTPUT	COVERAGE
Record counts	Implemented
Outcome percentages	Implemented
Grouped averages	Glucose · blood pressure · insulin
Age analysis	Exact-age grouping does not answer the intended age-band question
Next queries	BMI · glucose ranges · data quality · original-versus-cleaned comparisons

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REPORTING LAYER

POWER BI TURNS THE ANALYSIS INTO A PORTFOLIO-READY VIEW

CURRENT VIEW

Outcome distribution

Average glucose by Outcome

RECOMMENDED KPIS

Total Records

Positive Outcomes · Positive Outcome % · Average Glucose

PLANNED EXTENSIONS

More context

Age-group comparison · BMI-versus-glucose scatter plot · slicers



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THE HANDOFF IS THE RISK.

VALIDATION CHECK

THE KEY VALIDATION ISSUE IS IN THE DATA HANDOFF

Notebook upload

The notebook uploads `df`, the original dataframe, instead of `cleaned_df`.

Downstream risk

SQL and Power BI may therefore be using original zero values even after a cleaned dataset was created.

Required fix

Fix the upload, refresh Power BI, and reconcile dashboard metrics against the cleaned table.

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BEFORE PUBLISHING, TIGHTEN THE EVIDENCE CHAIN

01 Upload `cleaned_df` to the intended SQL table and refresh Power BI.

02 Correct age-band and percentage logic, including 100.0 division.

03 Recalculate the full correlation output from the final cleaned dataset.

04 Keep numeric medians as decimals unless rounding is intentional and documented.

PUBLISHING STANDARD

MAKE EVERY METRIC TRACEABLE.

From raw value to dashboard claim, the evidence chain should survive review.

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